

2D membrane vibrations (FD version)

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1 the problem

We solve the wave equation for Ψ its deviation from the equilibrium position at (x, y) point with damping

$$\frac{\partial^2 \Psi}{\partial t^2} = v^2 \nabla^2 \Psi - 2d \frac{\partial u}{\partial t} \quad (1)$$

for a square membrane $(x, y) \in (0, L) \times (0, L)$ using the finite element method. We set the speed of sound equal to $v = 1$, the size $L = 1$, and the damping constant $2d = 1$. In the central node of the mesh (c, c) we apply a periodic force $\sin \omega t$

We use a finite difference approach on a 2D mesh. Upon discretization of the derivatives with the usual approach for the second one and $\frac{\partial u}{\partial t} \simeq \frac{u(t) - u(t - \Delta t)}{\Delta t}$ we get

$$u_{ij}(t + \Delta t) = u_{ij}(t)[2 - 2d\Delta t] + u_{ij}(t - \Delta t)(2d\Delta t - 1) + \quad (2)$$

$$\frac{\Delta t^2 v^2}{\Delta x^2} (u_{i+1,j}(t) + u_{i-1,j}(t) + u_{i,j+1}(t) + u_{i,j-1}(t) - 4u_{i,j}(t)) \quad (3)$$

$$+ \delta_{i,c} \delta_{j,c} \Delta t^2 \sin(\omega t). \quad (4)$$

We start from a standing membrane $u(0) = u(\Delta t) = 0$. $\Delta x = L/N$, for $N \times N$ mesh. Lets set $v\Delta t = \Delta x/2$. We monitor the energy $E(t) = 0.5 \int_0^L dx \int_0^L dy \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right]$. We integrate the equation till $t = 80$.

2 Deliverables

- Lets set $\omega = \pi$. Does the motion obtained a periodic form?
- Lets us assume that the first 10 units of time is necessary for the membrane vibrations to reach their stationary form. We evaluate the average energy over the time interval $(10, 80)$. The average energy is $\langle E \rangle(\omega) = \frac{1}{70} \int_{10}^{80} dt E(t)$. Plot $\langle E \rangle(\omega)$ for $\omega = (1.5, 15)$.
- Determine the resonant frequencies.
- How the resonant peaks react to the change of the damping constant?